```
import pandas as pd
import seaborn as sns
import pandas as pd
# Convert the dataset to a pandas DataFrame (if it's not already)
# ISLP might already return a DataFrame, but this ensures it.
df_boston = pd.DataFrame(Boston)

# Save the DataFrame to a CSV file
df_boston.to_csv('Boston.csv')  # index=False prevents writing the DataFrame
index to the CSV

print("Boston dataset saved to Boston.csv")
Boston dataset saved to Boston.csv
College= load_data('College')
College.to_csv('College.csv')
```

(a) Use the pd.read_csv() function to read the data into Python. Call the loaded data college. Make sure that you have the directory set to the correct location for the data

```
college = pd.read_csv('College.csv')
```

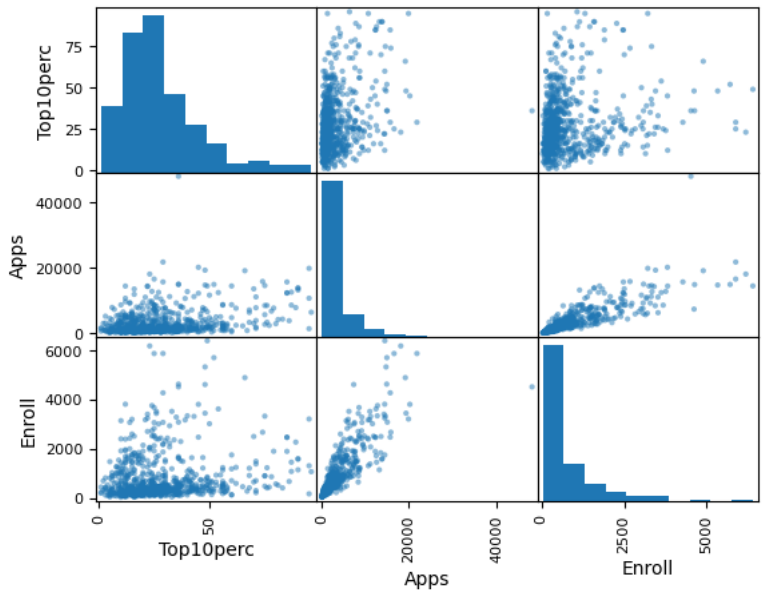
(b) Look at the data used in the notebook by creating and running a new cell with just the code college in it. You should notice that the first column is just the name of each university in a column named something like Unnamed: 0. We don't really want pandas to treat this as data. However, it may be handy to have these names for later. Try the following commands and similarly look at the resulting data frames:

```
college2 = pd.read_csv('College.csv', index_col=0)
college3 = college.rename({'Unnamed: 0': 'College'},
axis=1)
college3 = college3.set_index('College')
college = college3
college
```

c. Use the describe() method of to produce a numerical summary of the variables in the data set.

```
C = ['Top10perc','Apps','Enroll']
```

	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Under
count	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.00(
mean	3001.638353	2018.804376	779.972973	27.558559	55.796654	3699.907336	855.29{
std	3870.201484	2451.113971	929.176190	17.640364	19.804778	4850.420531	1522.43
min	81.000000	72.000000	35.000000	1.000000	9.000000	139.000000	1.00(
25%	776.000000	604.000000	242.000000	15.000000	41.000000	992.000000	95.00(
50%	1558.000000	1110.000000	434.000000	23.000000	54.000000	1707.000000	353.00(
75%	3624.000000	2424.000000	902.000000	35.000000	69.000000	4005.000000	967.00(
max	48094.000000	26330.000000	6392.000000	96.000000	100.000000	31643.000000	21836.00(

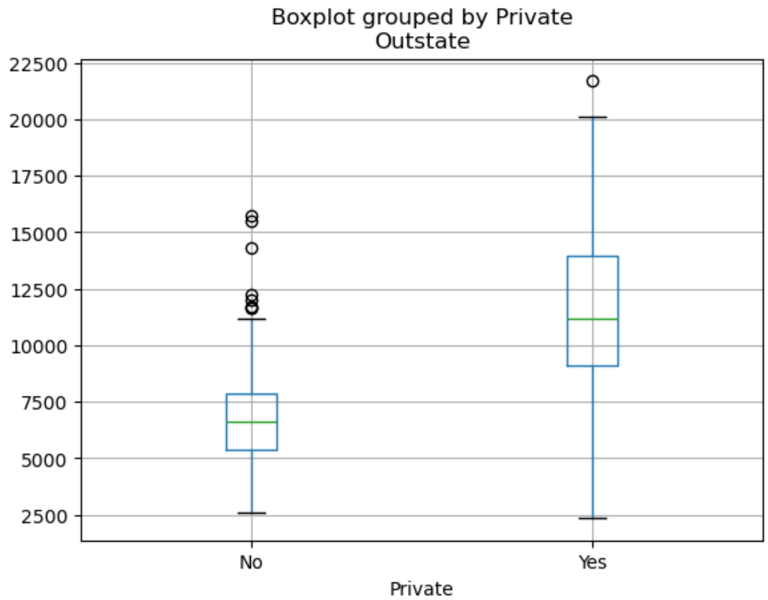


(d.)

	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Under
count	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.00
mean	3001.638353	2018.804376	779.972973	27.558559	55.796654	3699.907336	855.29
std	3870.201484	2451.113971	929.176190	17.640364	19.804778	4850.420531	1522.43
min	81.000000	72.000000	35.000000	1.000000	9.000000	139.000000	1.00
25%	776.000000	604.000000	242.000000	15.000000	41.000000	992.000000	95.00
50%	1558.000000	1110.000000	434.000000	23.000000	54.000000	1707.000000	353.00
75%	3624.000000	2424.000000	902.000000	35.000000	69.000000	4005.000000	967.00
max	48094.000000	26330.000000	6392.000000	96.000000	100.000000	31643.000000	21836.00

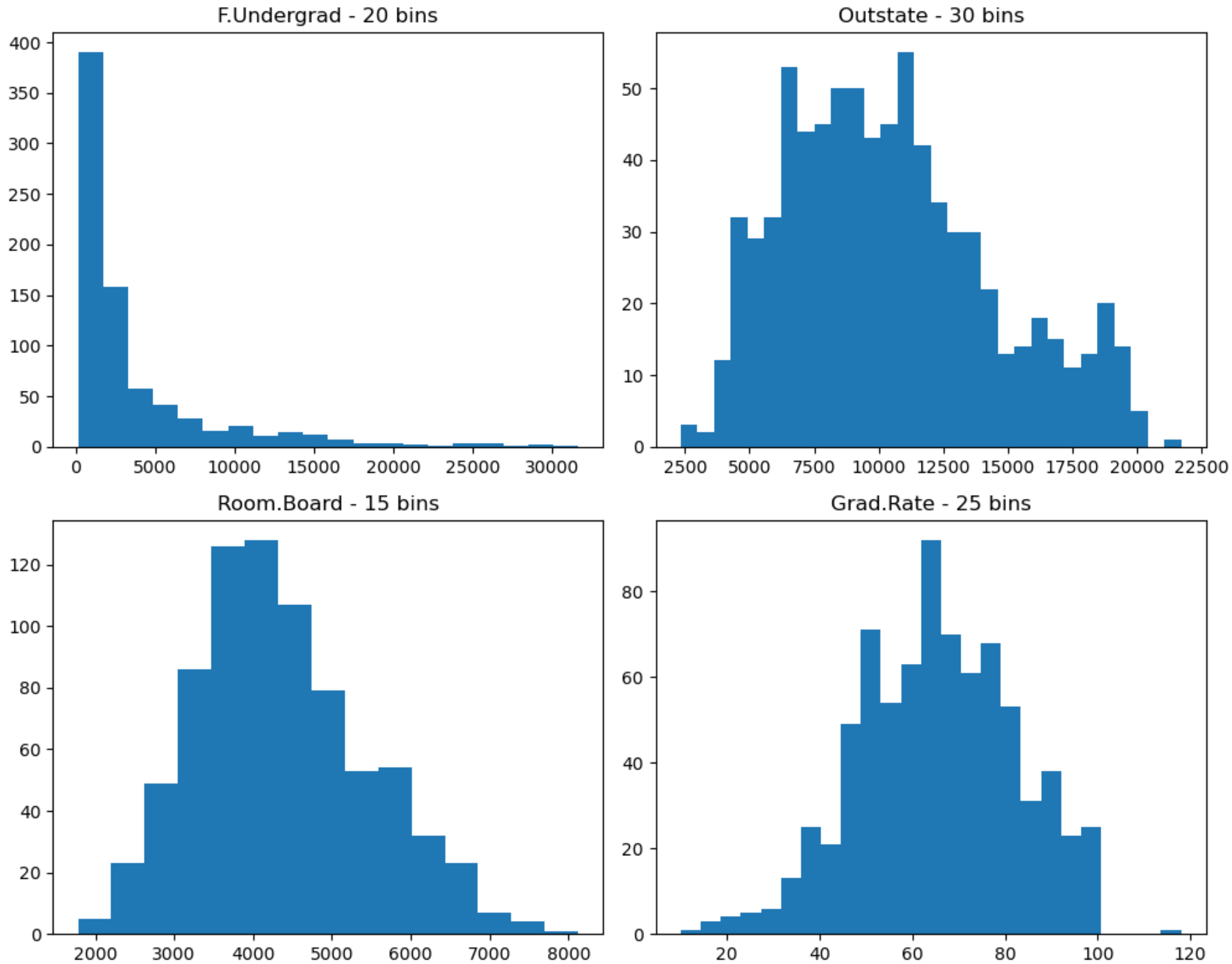
e) Use the boxplot() method of college to produce side-by-side boxplots of Outstate versus Private.

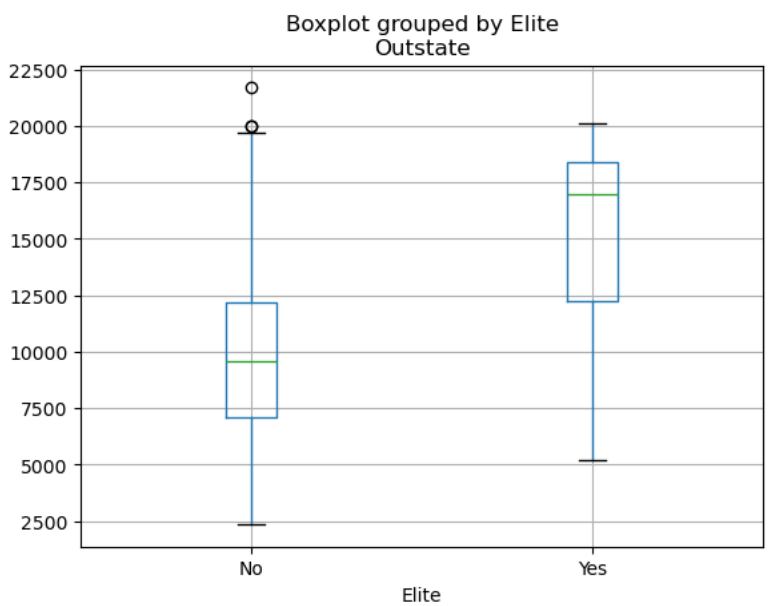
```
college.boxplot(column='Outstate', by='Private')
<Axes: title={'center': 'Outstate'}, xlabel='Private'>
```

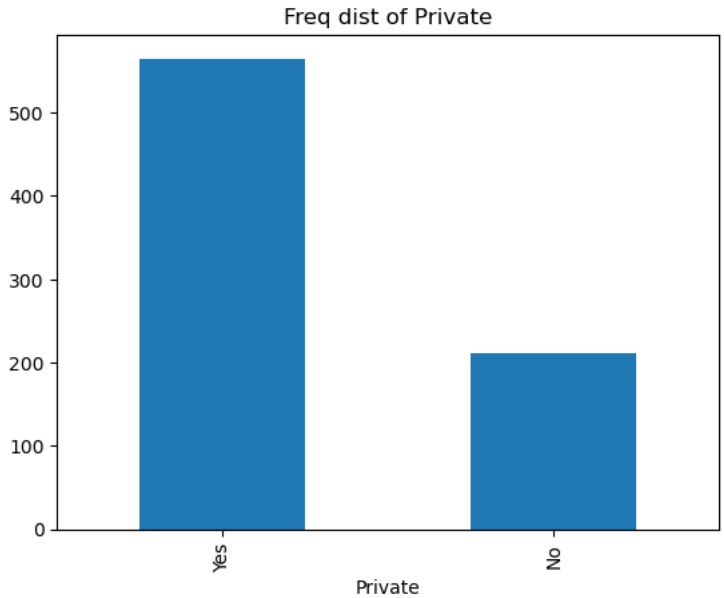


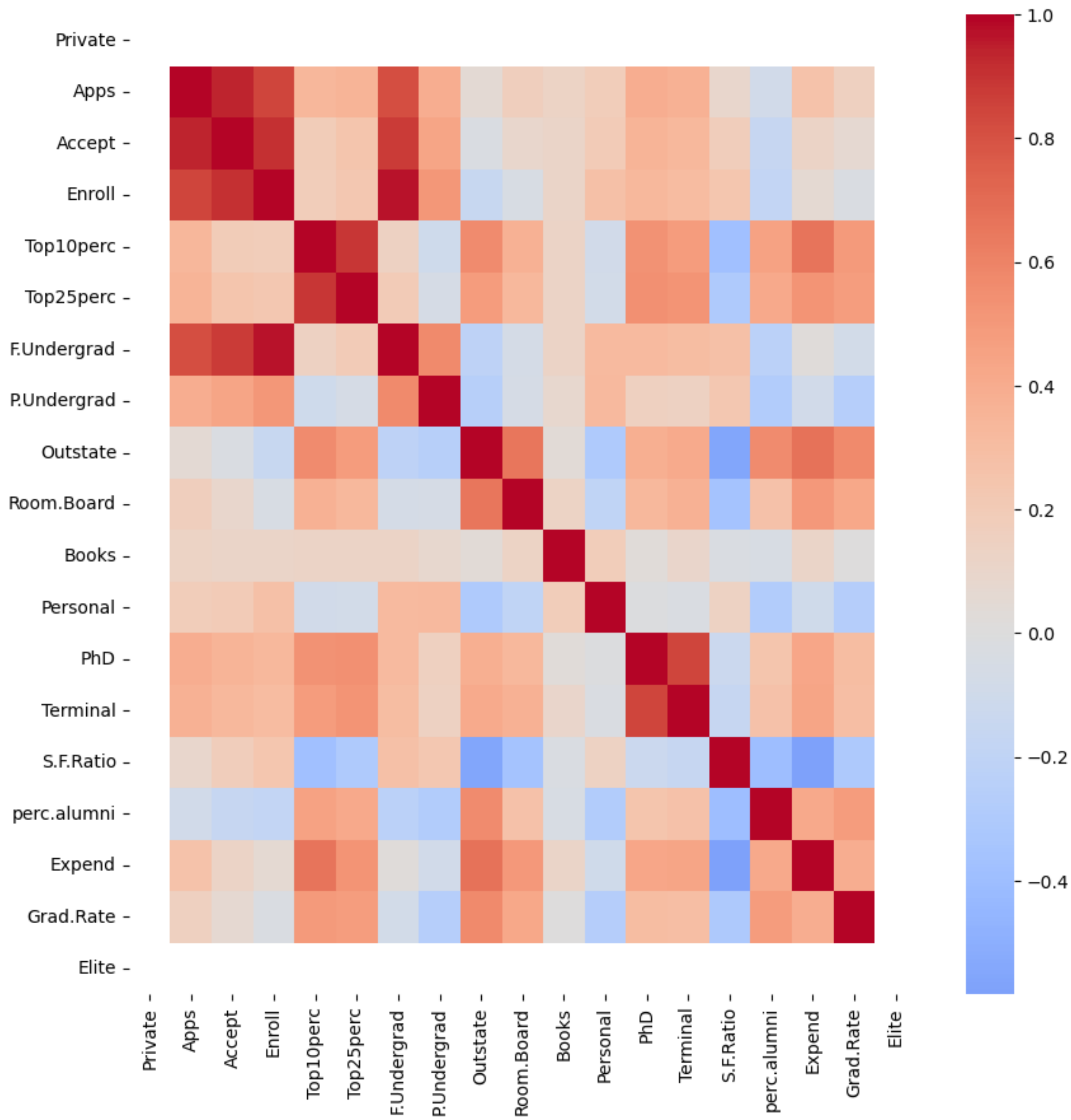
(f) Create a new qualitative variable, called Elite, by binning the Top10perc variable into two groups based on whether or not the proportion of students coming from the top 10% of their high school classes exceeds 50%.

```
# 建立 'Elite' 變數
college['Elite'] = ['Yes' if x > 50 else 'No' for x in college['Top10perc']]
# 查看各組校的數量
print(college['Elite'].value_counts())
Elite
No     699
Yes     78
Name: count, dtype: int64
# 繪製側並列箱型圖
college.boxplot(column='Outstate', by='Elite')
<Axes: title={'center': 'Outstate'}, xlabel='Elite'>
```


F.Undergrad - 20 bins
0
50
100
150
200
250
300
350
400
0
5000
10000
15000
20000
25000
30000
Outstate - 30 bins
0
10
20
30
40
50
2500
5000
7500
10000
12500
15000
17500
20000
22500
Room.Board - 15 bins
0
20
40
60
80
100
120
2000
3000
4000
5000
6000
7000
8000
Grad.Rate - 25 bins
0
20
40
60
80
20
40
60
80
100
120


Boxplot grouped by Elite
Outstate
2500
5000
7500
10000
12500
15000
17500
20000
22500
No
Yes
Elite


Freq dist of Private
500
400
300
200
100
0
Yes
No
Private


Private
Apps
Accept
Enroll
Top10perc
Top25perc
F.Undergrad
P.Undergrad
Outstate
Room.Board
Books
Personal
PhD
Terminal
S.F.Ratio
perc.alumni
Expend
Grad.Rate
Elite
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4